

WAZUH + GRAYLOG INTEGRATION

by Aftab Ahmed

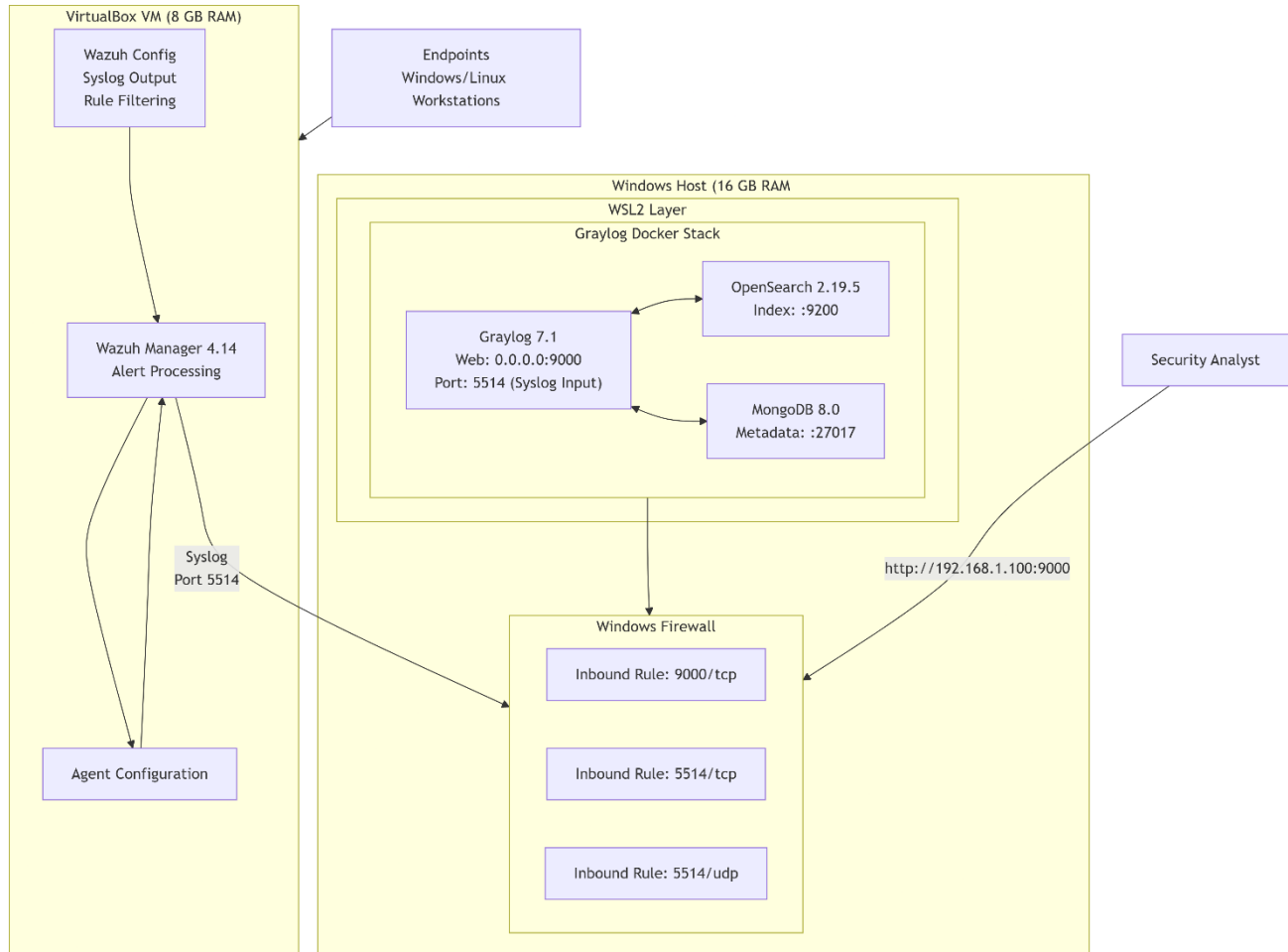


graylog

The freedom to build the perfect monitoring and visualization layer for your organization, without being locked into a single vendor's interface.

Architecture

Network Topology Diagram



Configure WSL2 Kernel Settings

Create the `.wslconfig` file in your Windows user profile:

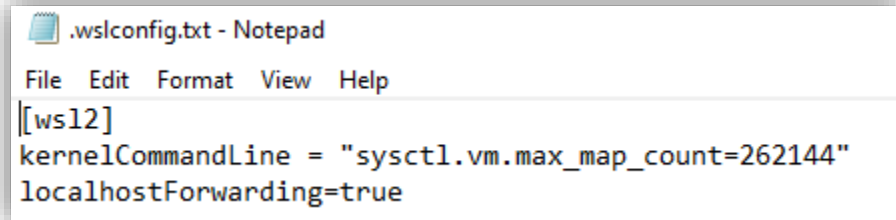
1. Open PowerShell:

- ❖ `cd $env:USERPROFILE`
- ❖ `notepad .wslconfig`

2. Add this content:

[wsl2]

- ❖ kernelCommandLine = "sysctl.vm.max_map_count=262144"
- ❖ localhostForwarding=true



3. Restart WSL:

- ❖ wsl --shutdown

4. Verify the setting:

- ❖ wsl -d docker-desktop -u root sysctl vm.max_map_count

```
PS C:\Windows\system32> wsl -l -v
  NAME                STATE      VERSION
* kali-linux          Stopped    2
  docker-desktop      Running    2
PS C:\Windows\system32> wsl -d docker-desktop -u root sysctl vm.max_map_count
vm.max_map_count = 262144
```

Prepare Windows Firewall Rules

Create inbound rules for Graylog ports:

PowerShell as Administrator:

Graylog Web UI (9000)

- ❖ New-NetFirewallRule -DisplayName "Graylog Web UI" -Direction Inbound -LocalPort 9000 -Protocol TCP -Action Allow

```

PS C:\Windows\system32> New-NetFirewallRule -DisplayName "Graylog Web UI" -Direction Inbound -LocalPort 9000 -Protocol TCP -Action Allow

Name                           : {c78efd4c-8bcf-450a-b08a-89603a543b33}
DisplayName                     : Graylog Web UI
Description                     :
DisplayGroup                    :
Group                           :
Enabled                         : True
Profile                         : Any
Platform                       : {}
Direction                      : Inbound
Action                         : Allow
EdgeTraversalPolicy             : Block
LooseSourceMapping              : False
LocalOnlyMapping                : False
Owner                           :
PrimaryStatus                   : OK
Status                         : The rule was parsed successfully from the store. (65536)
EnforcementStatus               : NotApplicable
PolicyStoreSource               : PersistentStore
PolicyStoreSourceType           : Local
RemoteDynamicKeywordAddresses  :
PolicyAppId                     :

```

Graylog Syslog Input (5514)

- ❖ New-NetFirewallRule -DisplayName "Graylog Syslog TCP" -Direction Inbound -LocalPort 5514 -Protocol TCP -Action Allow
- ❖ New-NetFirewallRule -DisplayName "Graylog Syslog UDP" -Direction Inbound -LocalPort 5514 -Protocol UDP -Action Allow

```

PS C:\Windows\system32> New-NetFirewallRule -DisplayName "Graylog Syslog TCP" -Direction Inbound -LocalPort 5514 -Protocol TCP -Action Allow

Name                           : {0295173e-a211-4036-8115-581af2f5ea35}
DisplayName                     : Graylog Syslog TCP
Description                     :
DisplayGroup                    :
Group                           :
Enabled                         : True
Profile                         : Any
Platform                       : {}
Direction                      : Inbound
Action                         : Allow
EdgeTraversalPolicy             : Block
LooseSourceMapping              : False
LocalOnlyMapping                : False
Owner                           :
PrimaryStatus                   : OK
Status                         : The rule was parsed successfully from the store. (65536)
EnforcementStatus               : NotApplicable
PolicyStoreSource               : PersistentStore
PolicyStoreSourceType           : Local

```

```

PS C:\Windows\system32> New-NetFirewallRule -DisplayName "Graylog Syslog UDP" -Direction Inbound -LocalPort 5514 -Protocol UDP -Action Allow

Name                           : {a8edf45c-f9fd-4a5f-8bc4-4bf25164cf37}
DisplayName                     : Graylog Syslog UDP
Description                     :
DisplayGroup                    :
Group                           :
Enabled                         : True
Profile                         : Any
Platform                       : {}
Direction                      : Inbound
Action                         : Allow
EdgeTraversalPolicy             : Block
LooseSourceMapping              : False
LocalOnlyMapping                : False
Owner                           :
PrimaryStatus                   : OK
Status                         : The rule was parsed successfully from the store. (65536)
EnforcementStatus               : NotApplicable
PolicyStoreSource               : PersistentStore
PolicyStoreSourceType           : Local

```

Graylog Installation

Step 1: Create Project Directory

In PowerShell

- ❖ `mkdir C:\graylog`
- ❖ `cd C:\graylog`

Step 2: Generate Security Secrets

- ❖ Generate Password Secret (96 characters)
- ❖ `$secret = -join ((48..57) + (65..90) + (97..122) | Get-Random -Count 96 | ForEach-Object {[char]$_})`
- ❖ `Write-Output "Secret: $secret"`
- ❖ Generate Root Password SHA2
- ❖ `$adminPassword = "StrongAdminPassword123!"`
- ❖ `$bytes = [System.Text.Encoding]::UTF8.GetBytes($adminPassword)`
- ❖ `$hash = [System.Security.Cryptography.SHA256]::Create().ComputeHash($bytes)`
- ❖ `$hashString = -join ($hash | ForEach-Object { $_.ToString("x2") })`
- ❖ `Write-Output "Password Hash: $hashString"`
- ❖ `Write-Output "Admin Password: $adminPassword"`

```
PS C:\graylog> $secret = -join ((48..57) + (65..90) + (97..122) | Get-Random -Count 96 | ForEach-Object {[char]$_})
PS C:\graylog> Write-Output "Secret: $secret"
Secret: 10OXDiUaZrhN7HeW9EMfT8IRbJgAYmu4FQ3p2GsBSdtyPV5o1wkxCKcLzqjv6
PS C:\graylog> $adminPassword = "1234"
>>
PS C:\graylog> $bytes = [System.Text.Encoding]::UTF8.GetBytes($adminPassword)
PS C:\graylog> $hash = [System.Security.Cryptography.SHA256]::Create().ComputeHash($bytes)
PS C:\graylog> $hashString = -join ($hash | ForEach-Object { $_.ToString("x2") })
PS C:\graylog> Write-Output "Password Hash: $hashString"
Password Hash: 03ac674216f3e15c761ee1a5e255f067953623c8b388b4459e13f978d7c846f4
PS C:\graylog> Write-Output "Admin Password: $adminPassword"
Admin Password: 1234
```

Step 3: Create docker-compose.yml

Create C:\graylog\docker-compose.yml:

```
C: > graylog > docker-compose.yml
1 services:
2   mongodb:
3     image: mongo:8.0
4     container_name: graylog-mongodb
```

```

volumes:
  - mongodb_data:/data/db
  - mongodb_config:/data/configdb
networks:
  - graylog-network
restart: unless-stopped
logging:
  driver: "json-file"
  options:
    max-size: "10m"
    max-file: "3"

opensearch:
  image: opensearchproject/opensearch:2.19.5
  container_name: graylog-opensearch
  environment:
    - cluster.name=graylog-cluster
    - node.name=opensearch-node1
    - discovery.type=single-node
    - bootstrap.memory_lock=true
    - "OPENSEARCH_JAVA_OPTS=-Xms1g -Xmx1g"
    - DISABLE_SECURITY_PLUGIN=true
    - DISABLE_INSTALL_DEMO_CONFIG=true

```

```

ulimits:
  memlock:
    soft: -1
    hard: -1
  nofile:
    soft: 65535
    hard: 65535
volumes:
  - opensearch_data:/usr/share/opensearch/data
networks:
  - graylog-network
restart: unless-stopped
logging:
  driver: "json-file"
  options:
    max-size: "10m"
    max-file: "3"

```

```

graylog:
  image: graylog/graylog:7.1
  container_name: graylog-server
  environment:
    # REPLACE THESE VALUES:
    - GRAYLOG_PASSWORD_SECRET=100XDiUaZrhN7Hew9EMfT8IRbJgAYmu4FQ3p2GsBSdtyPV5olwKxCKcLzqjv6
    - GRAYLOG_ROOT_PASSWORD_SHA2=03ac674216f3e15c761ee1a5e255f067953623c8b388b4459e13f978d7c846f4
    - GRAYLOG_HTTP_EXTERNAL_URI=http://192.168.1.100:9000/
    # END REPLACEMENTS

    - GRAYLOG_HTTP_BIND_ADDRESS=0.0.0.0:9000
    - GRAYLOG_MONGODB_URI=mongodb://mongodb:27017/graylog
    - GRAYLOG_ELASTICSEARCH_HOSTS=http://opensearch:9200
    - GRAYLOG_ROOT_TIMEZONE=UTC
    - GRAYLOG_SERVER_JAVA_OPTS=-Xms512m -Xmx512m -XX:+UseG1GC
  ports:
    - "9000:9000"      # Web UI
    - "5514:5514/tcp"  # Syslog TCP Input
    - "5514:5514/udp"  # Syslog UDP Input
  volumes:
    - graylog_data:/usr/share/graylog/data
  networks:
    - graylog-network
  restart: unless-stopped
  depends_on:
    - mongodb
    - opensearch

```

```

logging:
  driver: "json-file"
  options:
    max-size: "10m"
    max-file: "3"

volumes:
  mongodb_data:
  mongodb_config:
  opensearch_data:
  graylog_data:

networks:
  graylog-network:
    driver: bridge

```

Step 4: Deploy the Stack

In C:\graylog directory

❖ docker compose up -d

```

PS C:\graylog> docker compose up -d
[+] up 38/38
📦 Image opensearchproject/opensearch:2.19.5 Pulled
📦 Image graylog/graylog:7.1 Pulled
📦 Image mongo:8.0 Pulled
📦 Network graylog_graylog-network Created
📦 Volume graylog_graylog_data Created
📦 Volume graylog_opensearch_data Created
📦 Volume graylog_mongodb_config Created
📦 Volume graylog_mongodb_data Created
📦 Container graylog-opensearch Started
📦 Container graylog-mongodb Started
📦 Container graylog-server Started

```

Verify containers are running

❖ docker compose ps

```

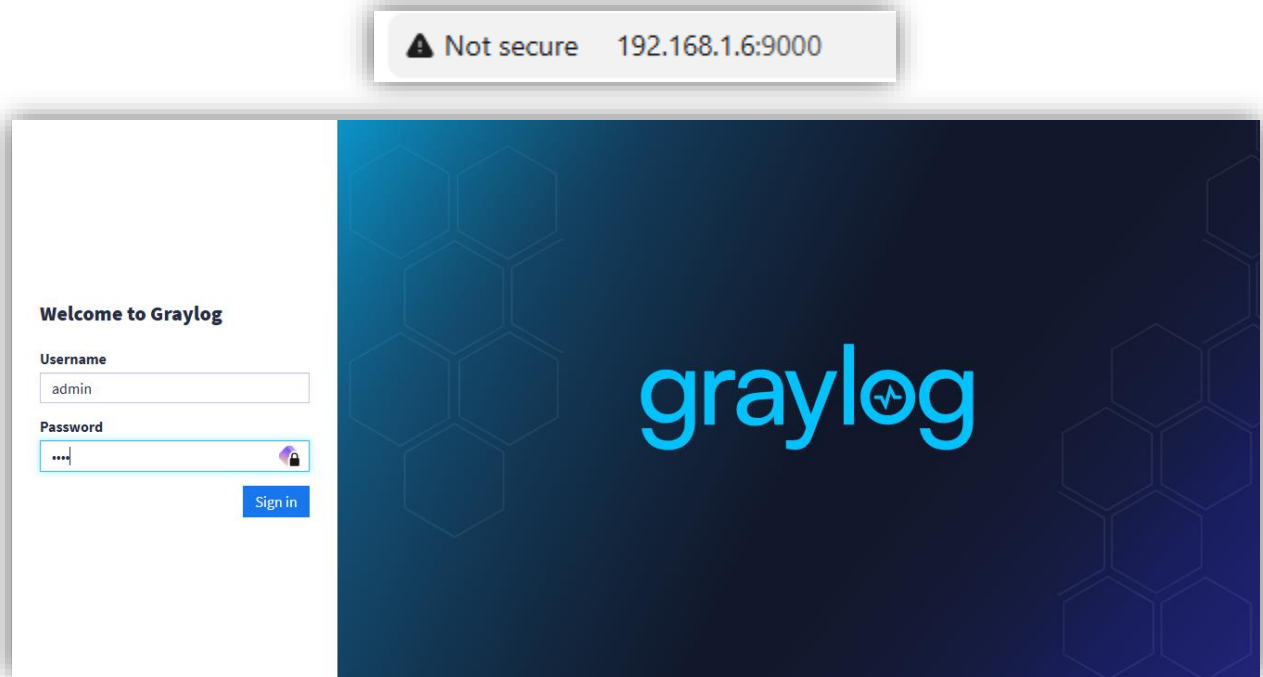
PS C:\graylog> docker compose ps
NAME                IMAGE                                COMMAND                                SERVICE    CREATED        STATUS
graylog-mongodb     mongo:8.0                          "docker-entrypoint.s..."            mongodb    About a minute Up
graylog-opensearch  opensearchproject/opensearch:2.19.5 "/opensearch-docker..."            opensearch About a minute Up
graylog-server      graylog/graylog:7.1                "tini -- /docker-ent..."            graylog    About a minute Up

```

NAME	IMAGE	COMMAND	SERVICE	CREATED	STATUS
graylog-mongodb	mongo:8.0	"docker-entrypoint.s..."	mongodb	About a minute ago	Up
graylog-opensearch	opensearchproject/opensearch:2.19.5	"/opensearch-docker..."	opensearch	About a minute ago	Up
graylog-server	graylog/graylog:7.1	"tini -- /docker-ent..."	graylog	About a minute ago	Up

Step 5: Access Graylog Web Interface

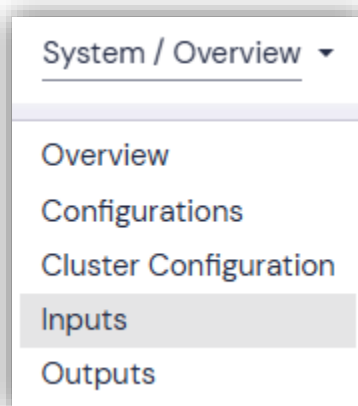
1. Open browser: `http://192.168.1.6:9000`
2. Username: admin
3. Password: The admin password you set



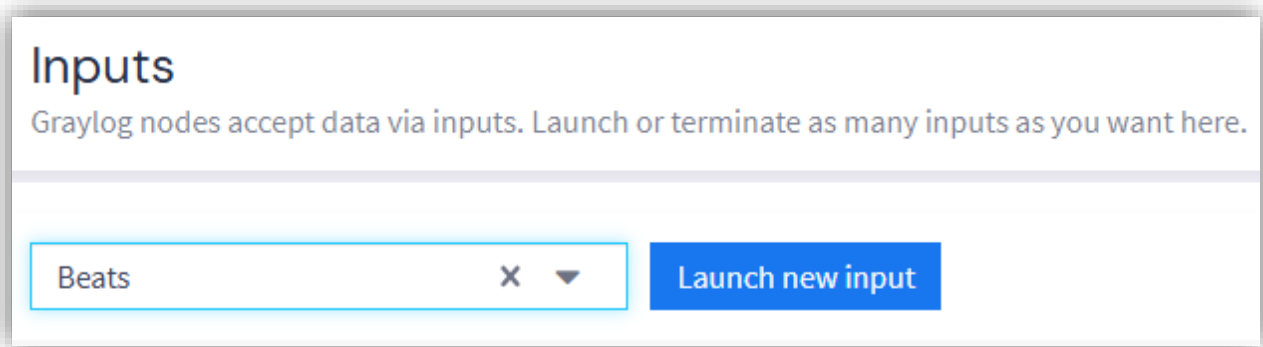
Step 6: Configure Graylog Beats Input

Create Syslog Input

1. Create the input:
 - Go to **System** → **Inputs**



- Click **Select Input** dropdown → **Beats**



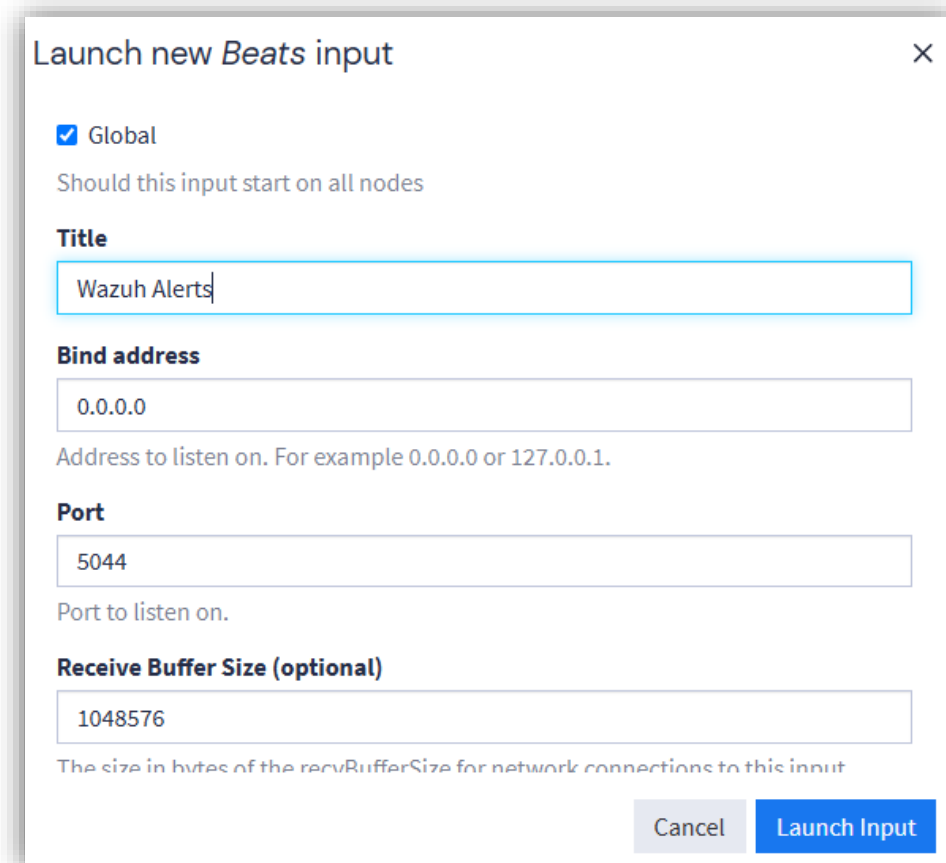
Inputs

Graylog nodes accept data via inputs. Launch or terminate as many inputs as you want here.

Beats X ▼

Launch new input

- Click **Launch new input**
- Configure:
 - **Title:** Wazuh Alerts
 - **Port:** 5044
 - **Bind address:** 0.0.0.0
 - **Override source:** Leave empty



Launch new *Beats* input X

☒ Global
Should this input start on all nodes

Title
Wazuh Alerts

Bind address
0.0.0.0
Address to listen on. For example 0.0.0.0 or 127.0.0.1.

Port
5044
Port to listen on.

Receive Buffer Size (optional)
1048576
The size in bytes of the `recvBufferSize` for network connections to this input

Cancel Launch Input

Create a Stream for Wazuh Alerts

1. Go to **Streams** → **Create stream**
2. **Title:** Wazuh Alerts
3. **Description:** Security alerts from Wazuh
4. Click **Save** then **Start stream**

The screenshot shows the 'Input Setup Wizard' dialog box with the 'Routing' tab selected. The wizard has three steps: 'Routing' (active), 'Launch', and 'Diagnosis'. Under 'Create new Stream', the 'Title' field is 'Wazuh Alerts' and the 'Description (Opt.)' field is 'Security alerts from Wazuh'. There are two checked checkboxes: 'Remove matches from 'Default Stream'' and 'Create a new pipeline for this stream'.

Input Setup Wizard

Routing Launch Diagnosis

Create new Stream

Title

Wazuh Alerts

A descriptive name of the new stream

Description (Opt.)

Security alerts from Wazuh

What kind of messages are routed into this stream?

☒ Remove matches from 'Default Stream'

Don't assign messages that match this stream to the 'Default Stream'.

☒ Create a new pipeline for this stream

The screenshot shows the 'Select Index Set' section of the 'Input Setup Wizard'. It has two tabs: 'Create a new Index Set' and 'Recommended!'. The 'Select Index Set' dropdown is set to 'Default index set'. Below it, a message states: 'Messages that match this stream will be written to the configured index set.' At the bottom right are 'Back' and 'Next' buttons.

Create a new Index Set Recommended!

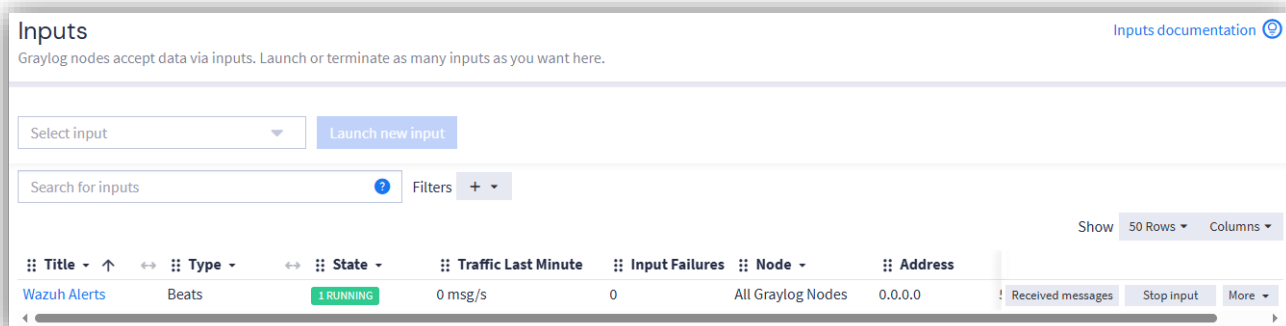
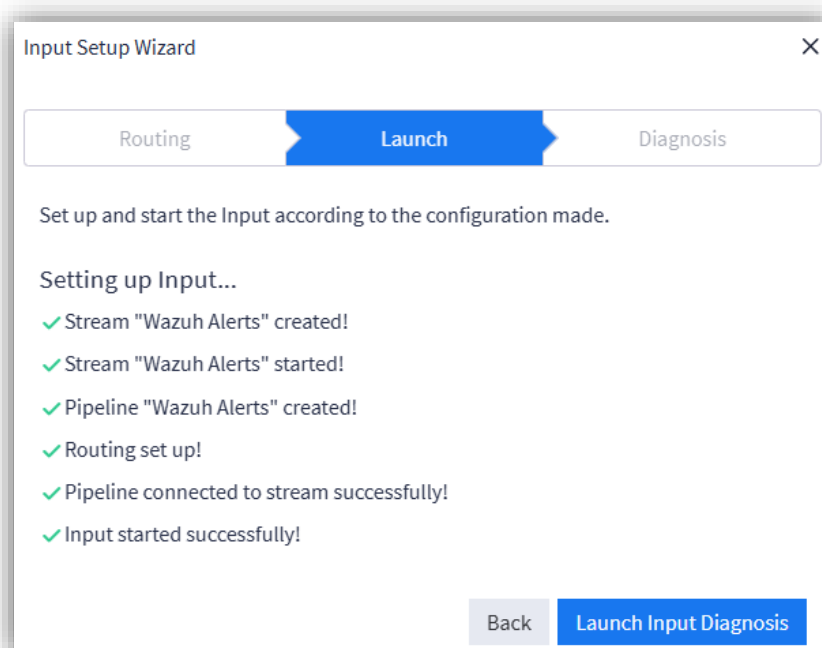
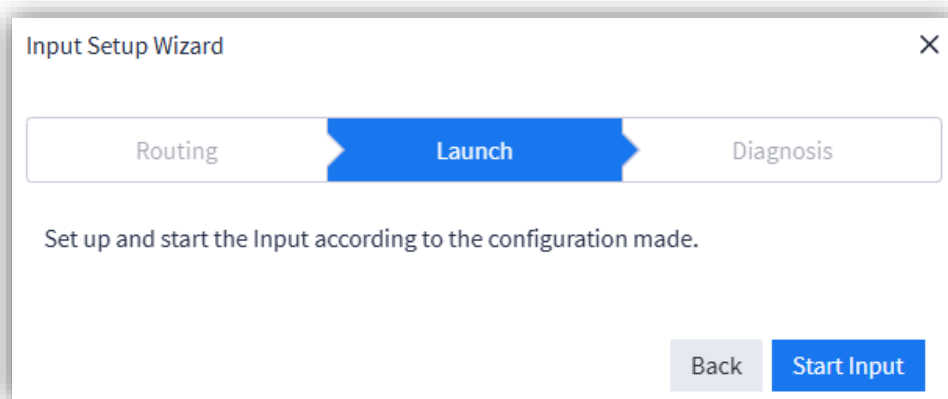
Select Index Set

Default index set

Messages that match this stream will be written to the configured index set.

Back Next

5. Click Start Input



Step 7: Configure Wazuh to Forward Alerts

Install and Configure Filebeat on Wazuh VM

Now we'll install Filebeat on Wazuh VM and configure it to read the alerts.json file.

1. Add Elastic's GPG key and repository

- ❖ `curl -s https://artifacts.elastic.co/GPG-KEY-elasticsearch | sudo apt-key add -`
- ❖ `echo "deb https://artifacts.elastic.co/packages/8.x/apt stable main" | sudo tee /etc/apt/sources.list.d/elastic-8.x.list`

2. Install Filebeat

- ❖ `sudo apt update`
- ❖ `sudo apt install filebeat -y`

```
wazuh@wazuh:~$ sudo apt install filebeat -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following package was automatically installed and is no longer required:
  libfwupd2
Use 'sudo apt autoremove' to remove it.
The following packages will be upgraded:
  filebeat
1 upgraded, 0 newly installed, 0 to remove and 6 not upgraded.
Need to get 70.4 MB of archives.
After this operation, 194 MB of additional disk space will be used.
Get:1 https://artifacts.elastic.co/packages/8.x/apt stable/main amd64 filebeat amd64 8.19.21 [70.4 MB]
Fetched 70.4 MB in 19min 14s (61.0 kB/s)
(Reading database ... 253701 files and directories currently installed.)
Preparing to unpack .../filebeat_8.19.21_amd64.deb ...
Unpacking filebeat (8.19.21) over (7.10.2-2) ...
Setting up filebeat (8.19.21) ...
```

Create the Filebeat Configuration :

- ❖ `sudo nano /etc/filebeat/filebeat.yml`

Replace the entire content with the following configuration. This is optimized for reading Wazuh's JSON alerts and forwarding them to Graylog :

```
GNU nano 7.2 /etc/filebeat/filebeat.yml *
# Filebeat configuration for Wazuh to Graylog
filebeat.inputs:
- type: log
  enabled: true
  paths:
    - /var/ossec/logs/alerts/alerts.json
  json.keys_under_root: true
  json.add_error_key: true
  json.message_key: full_log
  fields:
    source: wazuh
    fields_under_root: true

output.logstash:
  hosts: ["192.168.1.6:5044"]
```

```
processors:
- decode_json_fields:
  fields: ["message"]
  target: ""
  overwrite_keys: true

logging.metrics.enabled: false
```

Start and Verify Filebeat

Enable and start the Filebeat service, then verify it's running and can connect to Graylog :

Test the configuration

- ❖ `sudo filebeat test config`

Test the connection to Graylog

- ❖ `sudo filebeat test output`

```
wazuh@wazuh:~$ sudo filebeat test config
Config OK
wazuh@wazuh:~$ sudo filebeat test output
logstash: 192.168.1.6:5044...
connection...
  parse host... OK
  dns lookup... OK
  addresses: 192.168.1.6
  dial up... OK
TLS... WARN secure connection disabled
talk to server... OK
```

Start and enable Filebeat

- ❖ `sudo systemctl enable filebeat`
- ❖ `sudo systemctl start filebeat`
- ❖ `sudo systemctl status filebeat`

Generate Alerts to Test

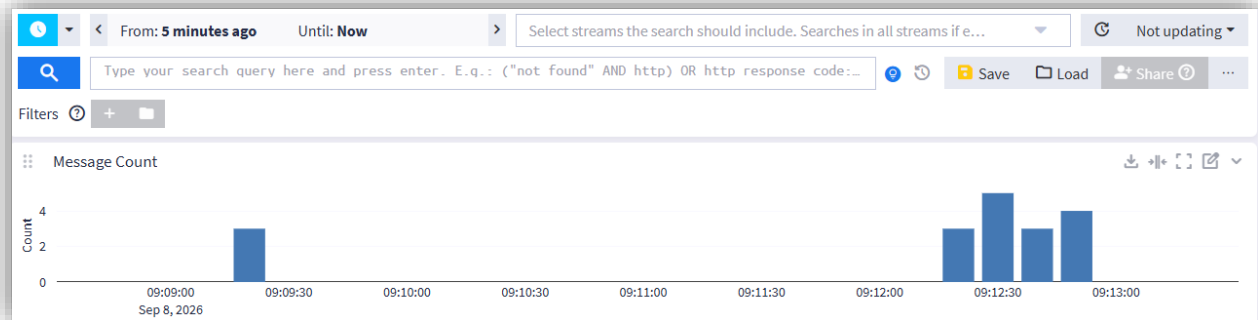
From your Wazuh VM, run to see new alerts appear in real-time:

Generate SSH failures

- ❖ `for i in {1..5}; do ssh invaliduser@localhost 2>/dev/null; done`

```
wazuh@wazuh:~$ for i in {1..5}; do ssh invaliduser@localhost 2>/dev/null; done
invaliduser@localhost's password:
invaliduser@localhost's password:
invaliduser@localhost's password:
invaliduser@localhost's password:
invaliduser@localhost's password:
invaliduser@localhost's password:
invaliduser@localhost's password:
invaliduser@localhost's password: 
```

In Graylog:



All Messages		
timestamp	source	
2026-09-08 09:12:59.340	wazuh	-
2026-09-08 09:12:52.336	wazuh	-
2026-09-08 09:12:51.335	wazuh	-
2026-09-08 09:12:51.331	wazuh	-
2026-09-08 09:12:48.329	wazuh	-
2026-09-08 09:12:47.327	wazuh	-

filebeat_data_srcip

127.0.0.1

filebeat_data_srcuser

invaliduser

filebeat_decoder_name

sshd

filebeat_decoder_parent

sshd

filebeat_ecs_version

8.0.0

filebeat_full_log

Sep 08 09:12:52 wazuh sshd[10344]: Failed password for invalid user invaliduser from 127.0.0.1 port 53636 ssh2

filebeat_rule_mitre_id

["T1110.001", "T1021.004"]

filebeat_rule_mitre_tactic

["Credential Access", "Lateral Movement"]

filebeat_rule_mitre_technique

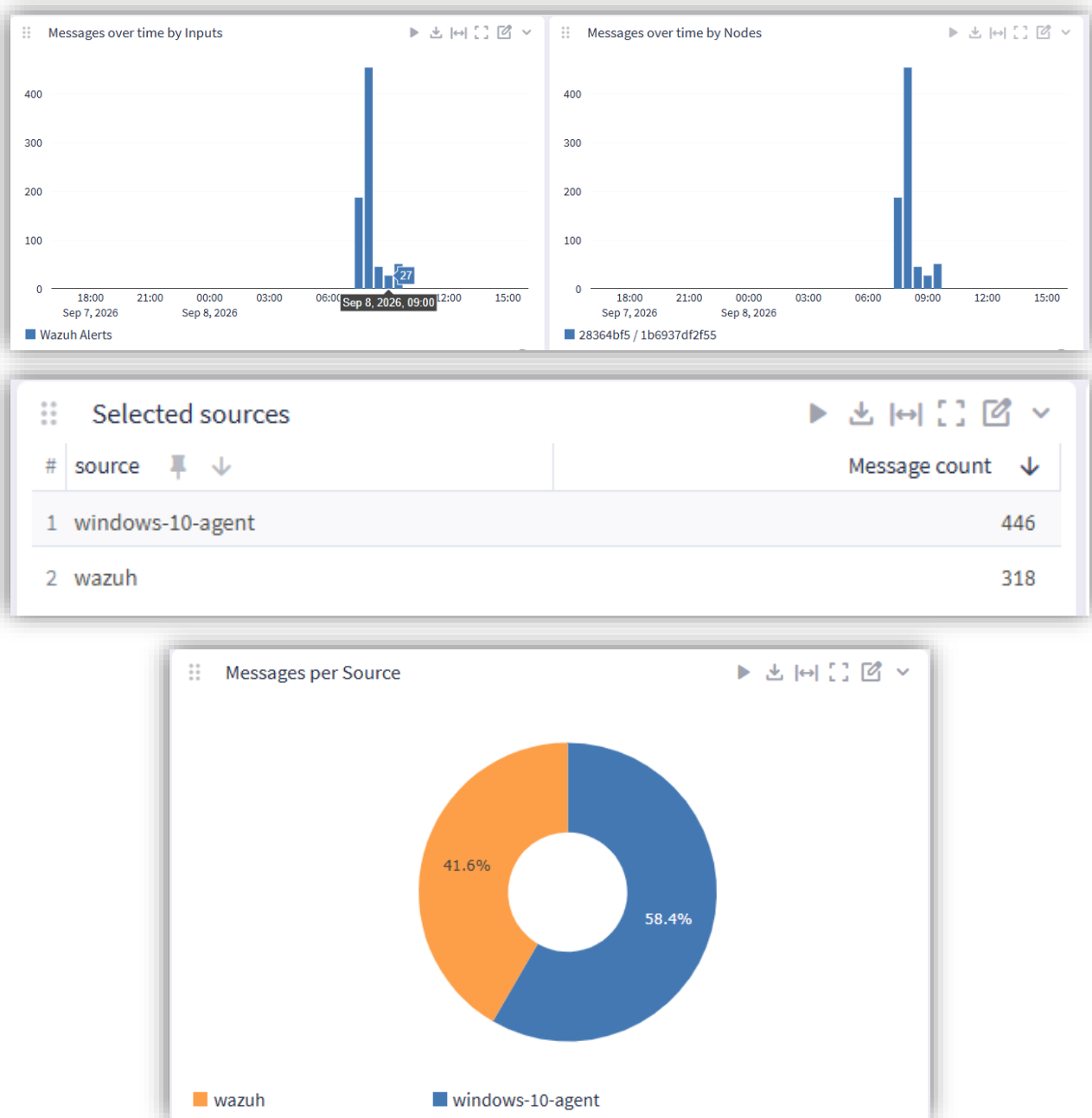
["Password Guessing", "SSH"]

Wazuh Logs are appearing successfully

Step 8: Create a Wazuh Alert Dashboard

In Graylog:

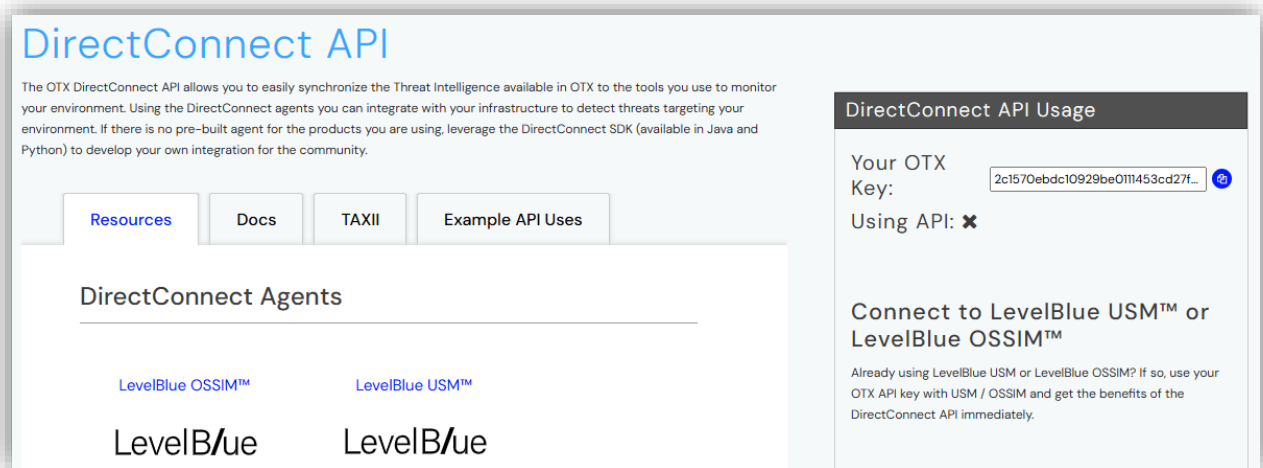
1. Go to **Dashboards** → **Create dashboard**
2. Add widgets:
 - **Alert count over time** (Bar chart)
 - **Alert level distribution** (Pie chart)



Step 9: AlienVault OTX Setup

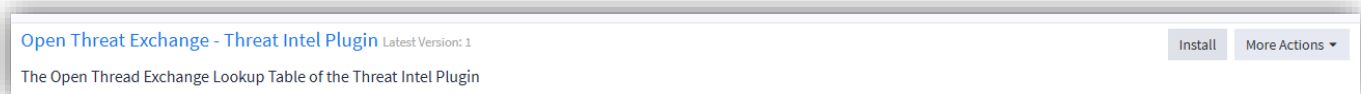
Get an OTX API Key

1. Go to <https://otx.alienvault.com/>
2. Create a free account
3. Navigate to your profile → **API Key**
4. Copy your personal API key

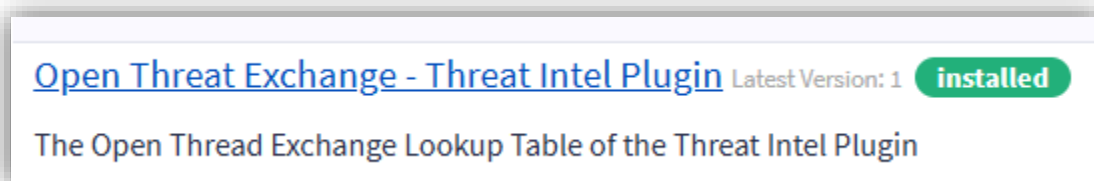


Install the OTX Content Pack in Graylog

1. In Graylog, go to **System** → **Content Packs**
2. Search for **Open Threat Exchange - Threat Intel Plugin**



3. Click the **Install** button



Create the OTX Lookup Table

1. Configure the **Data Adapter**:

- **Name:** otx-api-ip

Configure Adapter (Alienvault OTX API)

*** Title**

A short title for this data adapter.

Description

Data adapter description.

*** Name**

The name that is being used to refer to this data adapter. Must be unique.

Custom Error TTL ☐ 1 minutes ▾

Define a custom TTL for caching erroneous results. Otherwise the default of 5 seconds is used

Indicator

The OTX indicator type that should be used for lookups.

OTX API Key

Your OTX API key.

OTX API URL

URL of the OTX API server.

HTTP User-Agent

The User-Agent header that should be used for the HTTP request.

**HTTP Connect
Timeout**

HTTP connection timeout in milliseconds.

HTTP Write Timeout

HTTP write timeout in milliseconds.

HTTP Read Timeout

HTTP read timeout in milliseconds.

Test Known malicious IP (should show threat data)

Key

45.33.32.156

Key to look up a value for.

Look up

Lookup result

```
{
  "single_value": 3,
  "multi_value": {
    "whois": "http://whois.domaintools.com/45.33.32.156",
    "reputation": 0,
    "indicator": "45.33.32.156",
    "type": "IPv4",
    "type_title": "IPv4",
    "base_indicator": {
      "id": 3345785227,
      "indicator": "45.33.32.156",
      "type": "IPv4",
```

```
"pulse_info": {
  "count": 3,
  "pulses": [
    {
      "id": "670f97186e94777f0d520eec",
      "name": "C&C-IP-List",
      "description": "Command and Control List of IP",
      "modified": "2024-11-15T10:01:11.688000",
      "created": "2024-10-16T10:36:08.707000",
```

Similarly

Title	Description	Name	Throughput
Open Threat Exchange (OTX) - Domain	This is the data adapter for AlienVault's Open Threat Exchange platform, containing crowd-sourced IoCs (Indicators of Compromise). For more information see https://otx.alienvault.com. This adapter is used internally by Graylog's Threat Intel Plugin. Do not delete it manually.	otx-api-domain	0 lookups/s

Configure Adapter (Alienvault OTX API)

* Title

Open Threat Exchange (OTX) - Domain

A short title for this data adapter.

Description

This is the data adapter for AlienVault's Open Threat Exchange platform, cont

Data adapter description.

* Name

otx-api-domain

The name that is being used to refer to this data adapter. Must be unique.

Custom Error TTL

☐

1minutes

Define a custom TTL for caching erroneous results. Otherwise the default of 5 seconds is used

Indicator

Domain

The OTX indicator type that should be used for lookups.

Your OTX API key.

OTX API URL

URL of the OTX API server.

HTTP User-Agent

The User-Agent header that should be used for the HTTP request.

HTTP Connect Timeout

HTTP connection timeout in milliseconds.

HTTP Write Timeout

HTTP write timeout in milliseconds.

HTTP Read Timeout

HTTP read timeout in milliseconds.

Test Malicious Domain

Key

Key to look up a value for.

Look up

Lookup result

```
{
  "single_value": 12,
  "multi_value": {
    "sections": [
      "general",
      "geo",
      "url_list",
      "passive_dns",
      "malware",
      "whois",
      "http_scans"
    ]
  }
},
```

```
"pulse_info": {
  "count": 12,
  "pulses": [
    {
      "id": "69c08867316c564ade394c69",
      "name": "PhishDestroy - Content Active Threats (Live)",
      "description": "Live feed of phishing and crypto scam domains with ACTIVE m",
      "modified": "2026-09-08T16:06:48.073000",
      "created": "2026-03-23T00:25:09.116000",
      "tags": [
        "phishing",
        "crypto",
        "scam",
        "drainer",
        "fraud",
        "blocklist",
        "phishdestroy",
        "active",
        "content"
      ]
    },
  ],
}
```

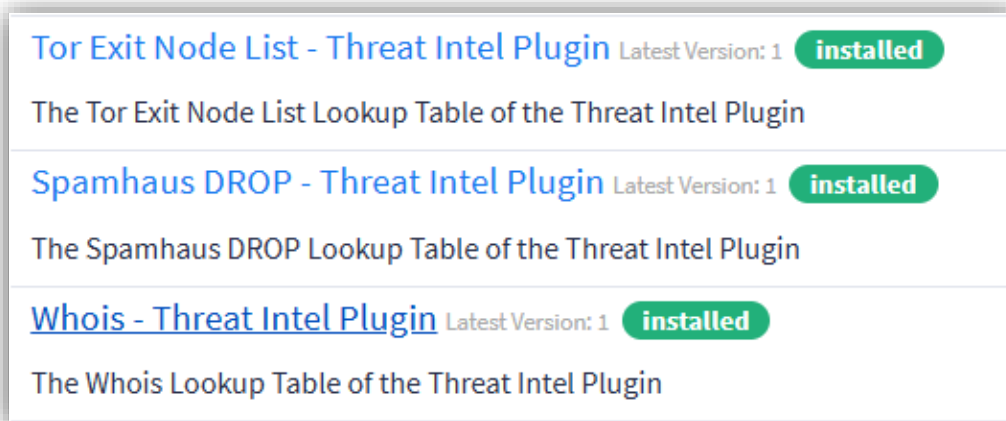
Other Available Threat Intel Packs

Pack	What It Does
Tor Exit Node List	Flags IPs known to be Tor exit nodes
Spamhaus DROP	Flags IPs from Spamhaus DROP/EDROP lists (known malicious networks)
Whois	Enriches IPs with WHOIS data (organization, country, etc.)

Install the Content Packs

For Each Pack:

1. Click "Install" on each pack



Enable Threat Intelligence Plugin

Before any of these work, you need to enable them in the plugin configuration:

1. Go to **System → Configurations**
2. From the left menu, select **Plugins**
3. Select **Threat Intelligence Lookup**

Threat Intelligence Lookup

Aws

Geo-Location Processor

Threat Intelligence Lookup Configuration

Configuration for threat intelligence lookup plugin.

Tor exit nodes:

Disabled

Spamhaus:

Disabled

Edit configuration

- Click Edit configuration
- Check the boxes

Update Threat Intelligence plugin Configuration

☒ Allow Tor exit node lookups?

Enable to include Tor exit node lookup in global pipeline function, disabling also stops refreshing the data.

☒ Allow Spamhaus DROP/EDROP lookups?

Enable to include Spamhaus lookup in global pipeline function, disabling also stops refreshing the data.

Cancel

Update configuration

Threat Intelligence Lookup Configuration

Configuration for threat intelligence lookup plugin.

Tor exit nodes:

Enabled

Spamhaus:

Enabled

Test 1: Whois

Test lookup

You can manually trigger the data adapter using this form. The data will be not cached.

Key

8.8.8.8

Key to look up a value for.

Lookup result

```
{
  "single_value": "Google LLC (GOGL)/US",
  "multi_value": {
    "organization": "Google LLC (GOGL)",
    "country_code": "US"
  },
  "string_list_value": null,
  "has_error": false,
  "ttl": 9223372036854775807
}
```

Test 2: Tor Exit Node

Official Tor exit node IP

Key

109.70.100.9

Key to look up a value for.

Look up

Lookup result

```
{
  "single_value": "8D345A709C073C117768A1F210CD04BF12234AEC, A93D8E7E1E1EA88F5BBC819883C9364154317DDC",
  "multi_value": {
    "node_ids": [
      "8D345A709C073C117768A1F210CD04BF12234AEC",
      "A93D8E7E1E1EA88F5BBC819883C9364154317DDC",
      "A4F0F516C83DE11B290384B9B4A4C928A78ED3A5",
      "D4AB390149B8EA38A9E33058E5624207893242BD",
      "37C25C1E9CA9F4872D536D084EDDA57F66B43435",
      "8A3D2DF07B66DDF8F81D2366C2C9D06209CFAF7",
      "C7FE633139E83E5E3653057A992DE93D74479ECC",
      "10E9A196646DA911893878A78368D67C7B46330A"
    ]
  },
  "string_list_value": null,
  "has_error": false,
  "ttl": 9223372036854775807
}
```

Benefits of the Wazuh + Graylog Integration

- **Wazuh: The Dedicated Detective**

By offloading the heavy lifting of log parsing and management, Wazuh is free to focus all its energy on its core strengths: real-time **threat detection**, **file integrity monitoring (FIM)**, and **vulnerability assessment**. This means you get a more responsive and effective endpoint security solution.

- **Graylog: The Central Nervous System**

Graylog acts as the brain of your security operations, centralizing and normalizing data from Wazuh and virtually any other source—be it firewalls, cloud services, or custom applications. Its powerful search and analysis capabilities allow you to investigate security events with surgical precision. In short, **Wazuh detects**, **Graylog enriches and analyzes**.

The Result: A highly optimized, specialized workflow where each component excels at its assigned role, leading to faster, more reliable threat identification.

Benefits of Threat Intelligence Enrichment

Feed	Key Benefit	Provides Context For
Tor Exit Node List	Proactive detection of Tor-related activity	Anonymity networks, potential malicious scans
Spamhaus (E)DROP	Blocking known malicious networks	Spamming, malware, botnet command and control
Whois for IPs	Detailed owner and location data	Incident response, asset identification
AlienVault OTX	Real-time IOC matching and threat naming	Phishing, malware C2, brute-force campaigns

End of Write-up